

AWS-Based Data Science Group Project

Customer Churn Prediction and Analytics Solution

Group Size - 4 Students

Duration - 2 Weeks

Project Scenario

A telecommunications company is experiencing customer churn and would like to better understand why customers leave and identify customers who may be at risk of leaving.

Your group is required to design and implement a cloud-based Data Science solution using AWS to analyse customer data, develop a predictive model, and produce useful business insights.

Project Requirements

Your solution should be able to

- Store the original customer dataset in the cloud.
- Organize raw, processed and output data appropriately.
- Automatically detect or process newly uploaded datasets.
- Perform basic validation of uploaded data.
- Maintain logs that can be used to monitor or troubleshoot the processing workflow.
- Perform Exploratory Data Analysis on the customer dataset.
- Identify missing values, incorrect data types and other data-quality issues.
- Produce at least four meaningful visualizations.
- Perform suitable data preprocessing.
- Develop at least two machine-learning classification models.
- Evaluate and compare the models using suitable classification metrics.
- Identify the customer characteristics that appear to be associated with churn.
- Store processed datasets and important outputs in the cloud.
- Perform at least five meaningful SQL-based analytical queries on the processed data.
- Provide at least three business recommendations based on the analysis.
- Apply appropriate security practices to protect cloud resources and credentials.
- Monitor cloud resource usage and avoid unnecessary costs.

Students should select suitable **AWS services and architecture** to satisfy these requirements.

Submission Requirements

Each group should submit -

1. Technical Report – approximately 6–8 pages
2. Data Science notebook/source code
3. Cloud-related source code or configuration used in the project

4. SQL queries used for the analysis
5. AWS Architecture Diagram
6. GitHub Repository Link
7. Individual Contribution Table
8. Short Group Demonstration

Important Notes

Students are expected to use AWS Free Tier / Free Plan resources wherever possible.

Each group must justify the AWS services selected for the implementation.